

DAY-3

1. List 5 common BIOS or POST error messages you might see when starting a PC and write what hardware or configuration issue each one usually points to.

BIOS/POST Error Message	Hardware or Configuration Issue
CMOS Checksum Error	The CMOS battery is weak or BIOS settings have been reset.
Keyboard Not Detected	The keyboard is not connected or is faulty.
No Boot Device Found	The hard disk/SSD is not detected or does not have an operating system.
Memory Error (RAM Error)	The RAM is loose, faulty, or not installed correctly.
CPU Fan Error	The CPU fan is not connected, not spinning, or has failed.

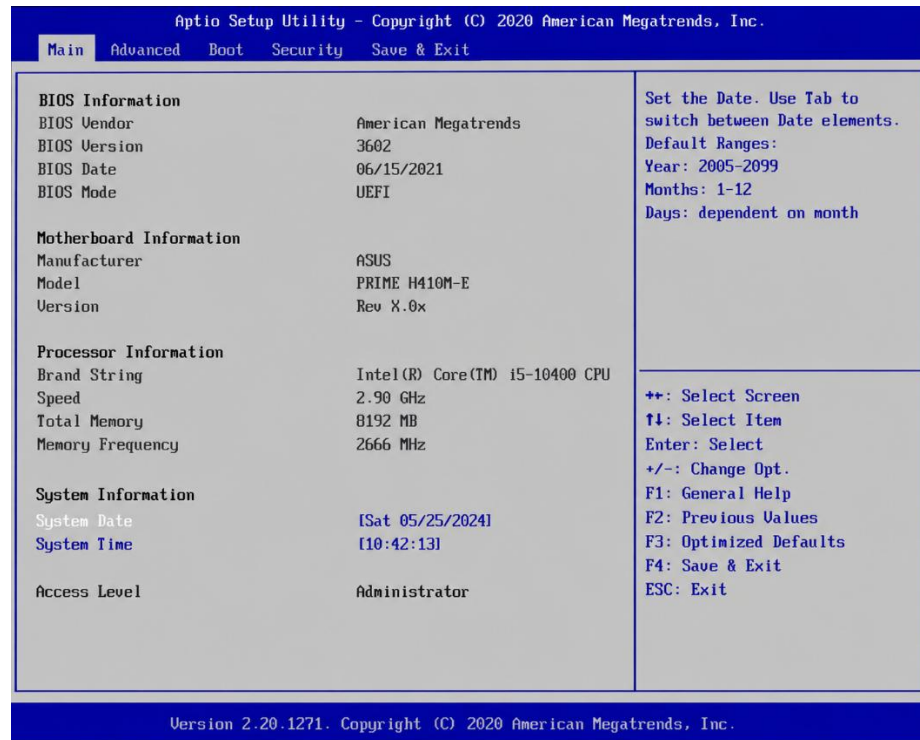
2. Given this scenario: On boot, a PC beeps 3 times in a repeating pattern and does not display anything on the screen. Research and write what this POST beep code typically means for an AMI BIOS, and suggest two troubleshooting steps.

A 3-beep repeating pattern on an AMI BIOS usually indicates a RAM (memory) problem. The RAM may be loose, faulty, or not installed correctly, which prevents the computer from starting and displaying anything on the screen.

Troubleshooting Steps

1. Remove and reinsert the RAM properly, then restart the computer.
2. Test one RAM stick at a time or try a different RAM slot to check if the RAM or slot is faulty.

3. Access the BIOS setup utility on any available PC or virtual machine, take a screenshot of the main BIOS screen, and identify the motherboard manufacturer and BIOS version.



- Motherboard manufacturer:- ASUS
- BIOS version:- 3602 (AMI BIOS)

4. You are given a screenshot of a POST error message that says 'CMOS checksum error – Defaults loaded.' Explain what this means and describe how you would resolve this issue in a real system.

Hint: Think about the CMOS battery and BIOS settings reset.

What does it mean?

This error means the BIOS settings have been reset because the CMOS battery is weak or dead, or the BIOS settings were lost. The computer loads the default BIOS settings so it can start.

Steps to fix it:-

1. Restart the computer and enter the BIOS Setup by pressing Delete or F2.
2. Check and correct the date and time if they are wrong.
3. Save the BIOS settings and restart the computer.
4. If the error appears again, replace the CMOS battery on the motherboard.
5. Restart the computer and set the BIOS settings again if needed.